

# Data Dictionary

## Group-Based Trajectory Modeling, Two Continuous Outcomes (Censored Normal)

File: continuous\_2outcomes.sas

Last updated: July 21, 2026

Package: Traj2 (released core)

### 1. Overview

This file implements group-based trajectory modeling (GBTM) for one or two continuous outcomes with bounded support, using a censored-normal (Tobit) likelihood fit by maximum likelihood in PROC NLMIXED. It supports single-outcome fits and a joint two-outcome (multi-trajectory) model that shares one latent-class structure. The joint model uses a warm-start pipeline: each outcome is fit separately first, and those estimates seed the joint fit.

### 2. Datasets

| Name                | Description                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| BASE_FILE_SRS       | Wide input table: one row per subject, repeated measures per outcome across T time points, plus time-index and cap columns. |
| sim_long / sim_wide | Long and wide outputs of the optional simulator (%sim_data_cont).                                                           |
| nlm_fix_T1&class    | Parameter estimates, single-outcome fit for outcome 1.                                                                      |
| nlm_fix_T2&class    | Parameter estimates, single-outcome fit for outcome 2.                                                                      |
| nlm_2y_starting     | Stacked single-outcome estimates (alpha rows dropped) that seed the joint fit.                                              |
| nlm_fix_T1_T2&class | Parameter estimates, joint two-outcome fit.                                                                                 |
| parameter           | Estimates reshaped to one row for prediction/plotting.                                                                      |
| data_pred           | BASE_FILE_SRS crossed with parameters for posterior computation.                                                            |
| pred_membership_y   | Posterior class-membership probabilities per subject.                                                                       |
| avg_y1 / avg_y2     | Posterior-weighted observed means per class (via PROC IML).                                                                 |
| pred / data_plot    | Predicted and merged predicted-vs-observed data for plotting.                                                               |
| avg_membership      | Averaged posterior class-membership probabilities.                                                                          |

### 3. Global Macro Variables (Step 0)

| Name    | Example | Description                                            |
|---------|---------|--------------------------------------------------------|
| USE_SIM | 1       | 1 = run the simulator; 0 = use your own BASE_FILE_SRS. |

| Name        | Example      | Description                                                  |
|-------------|--------------|--------------------------------------------------------------|
| T           | 12           | Number of time points.                                       |
| class       | 5            | Number of latent classes.                                    |
| order_model | 3            | Polynomial order (1=linear, 2=quadratic, 3=cubic).           |
| equal       | T            | T = equal residual sigma across classes; F = class-specific. |
| max_values  | 10 10 ... 10 | Censoring caps, one per time point (length T).               |
| y1vars      | QHH1-QHH12   | Outcome 1 repeated-measure variables.                        |
| y2vars      | QINP1-QINP12 | Outcome 2 repeated-measure variables.                        |
| idvar       | BENE_ID      | Subject identifier variable.                                 |
| tvars       | quar1-quar12 | Time-index variables (values 1..T).                          |

## 4. Key Variables

| Name                     | Type  | Description                                               |
|--------------------------|-------|-----------------------------------------------------------|
| BENE_ID                  | Num   | Subject identifier; carried to output for linkage.        |
| QHH1-QHH12 (y1vars)      | Num   | Outcome 1 repeated measures, one column per time point.   |
| QINP1-QINP12 (y2vars)    | Num   | Outcome 2 repeated measures.                              |
| quar1-quar12 (tvars)     | Num   | Time indices, values 1..T.                                |
| CAP1-CAP&T               | Num   | Per-time censoring caps loaded from max_values.           |
| X[T]                     | Array | Time predictor inside the model.                          |
| Y1[T], Y2[T]             | Array | Outcome arrays inside the model.                          |
| alpha0_[class]           | Num   | Class-membership intercept (class A fixed to 0).          |
| beta[outcome]_[class][k] | Num   | Degree-k polynomial coefficient, given outcome and class. |
| sigma[outcome]_[class]   | Num   | Residual SD; class suffix omitted when equal_sigma=T.     |
| mu[class][outcome][T]    | Array | Class- and outcome-specific latent mean over time.        |
| e[outcome]_[class]       | Num   | Residual (observed - predicted).                          |
| pie_[class]              | Num   | Marginal class-membership probability.                    |
| pinumer_[class] / pideno | Num   | Numerator / denominator of the class-probability softmax. |
| post_[class]             | Num   | Posterior probability of class membership per subject.    |

| Name                     | Type | Description                                                |
|--------------------------|------|------------------------------------------------------------|
| <code>ll_latclass</code> | Num  | Log-likelihood contribution passed to the model statement. |

## 5. Macros

| Name                                       | Purpose                                                       |
|--------------------------------------------|---------------------------------------------------------------|
| <code>%sim_data_cont</code>                | Optional simulator: two censored-normal outcomes.             |
| <code>%add_caps</code>                     | Loads CAP1-CAP&T from the max_values list.                    |
| <code>%build_base_file_srs</code>          | Assembles BASE_FILE_SRS (simulated or user-supplied).         |
| <code>%starting_value_alpha</code>         | Neutral starting values for mixing intercepts (A fixed to 0). |
| <code>%starting_value_beta_sigma</code>    | Starting values for polynomial coefficients and sigma.        |
| <code>%bounds_alpha / %bounds_sigma</code> | Parameter bounds for stable estimation.                       |
| <code>%initiation_universal</code>         | Declares shared time/outcome arrays.                          |
| <code>%initiation</code>                   | Declares per-class, per-outcome arrays.                       |
| <code>%model</code>                        | Builds the polynomial mean for each class.                    |
| <code>%residual</code>                     | Computes residuals per class.                                 |
| <code>%float_control</code>                | Caps extreme residuals for numerical stability.               |
| <code>%Prob_cnorm</code>                   | Censored-normal log-density contribution per time point.      |
| <code>%Class_Membership</code>             | Softmax class-membership probabilities.                       |
| <code>%LogLike / %LogLike_multi</code>     | Single-outcome / joint mixture log-likelihood.                |
| <code>%Posterior</code>                    | Posterior class probabilities per subject.                    |
| <code>%Pred_cnorm</code>                   | Censored-mean prediction per class.                           |
| <code>%plot_prep</code>                    | Builds predicted-vs-observed plots and posterior summary.     |
| <code>%nlmixed_1</code>                    | Fits a single-outcome model.                                  |
| <code>%nlmixed_MultiTraj</code>            | Fits the joint two-outcome model.                             |

## 6. Notes and Assumptions

- Missing time points are allowed (.) and skipped in the likelihood, valid under missing-at-random.
- Censoring is recognized by exact equality of an outcome to its cap; the cap vector must match the true measurement limits: validate against the sample max.

- Class labels A, B, C, ... come from %let class\_names; changing the count must be reflected in the class= / LC= arguments.
- equal\_sigma=T estimates one sigma across classes; F estimates one per class.
- The joint fit depends on starting values: use the warm-start pipeline (single fits → stack → joint) to reduce non-convergence.